Data Safari #4

Week 13 · Replicating a published figure

The last Data Safari of the semester — same format as Data Safari #1. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas alongside this week's regular assignment, before Week 14's lecture (Monday). See the syllabus for the late policy.

Tip: now that you've covered mixed-effects models, a great figure to look for is one showing **repeated measurements over time or across sites**, with individual trajectories (often shown as thin lines) and an overall group trend layered on top — the kind of data a mixed model would be built to analyze.

Instructions

1. **Find a paper** with a figure of this kind and publicly available data.

Full citation:

2. **Recreate the figure**, working from the actual data file.
3. **Reflect**: how easy was the data to find and use? How closely does your figure match? What would you change about the paper's data documentation?

Bonus (optional)

If you can identify the grouping structure (subject, site, etc.) in the data, try fitting a simple mixed model and see whether the random-effect variance looks large relative to the fixed effect — does that match the visual spread you see in the figure? Use `avg_slopes()` or `avg_comparisons()` from `marginaleffects` to report the population-level effect with a confidence interval.